








Blockchain Applications

Lecture 3: Crypto Indices: CRIX, VCRIX and Investible Index Funds



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Opening: Measuring the Crypto Market with an Index

"While most CC indices look similar at the structural level, their design choices create fundamentally different exposures."
— Găman, Będowska-Sójka, Pele & Härdle (2026), White paper on cryptocurrency indices (CC: cryptocurrency)

- The Royalton CRIX (CRypto IndeX) rose from 1 960 on 16 March 2018 to a record 21 523.34 on 6 October 2025, then fell 56% to 30 June 2026
- CRIX is one of nine types of crypto indices: market, sector, price, futures, volatility, yield, smart-beta, fundamentals and sentiment Găman, Będowska-Sójka, Pele & Härdle (2026)
- Our own replica of CRIX, built on ten coins, tracks the published index with a correlation of 0.990 of daily returns
  - Bitcoin alone: 0.971; the remaining tracking error of the replica is 9.3% a year
- VCRIX, the volatility index of CRIX, peaked at 1 215.57 the day after the crash of 12 March 2020

▸ Central chart: The Replica against the Real CRIX

Today's Question and Route

- **Question**: how do we summarise a market of thousands of coins in one number that an investor can actually buy?
  - a fund needs a benchmark; an exchange-traded product needs an index it can replicate at low cost
- **Route**
  - why an index: concentration and survivorship bias
  - how CRIX is built: Laspeyres formula, divisor, liquidity rule, number of constituents by an information criterion
  - our own replication on real data, compared with the published CRIX
  - the real CRIX 2017–2026 and VCRIX, its volatility index
  - from index to fund: tracking difference and tracking error of a spot Bitcoin fund
  - beyond CRIX: nine types of crypto indices, their formulas, two of them rebuilt on our data
- Next lecture (Lecture 4): volatility models and tail risk; the generalised autoregressive conditional heteroskedasticity (GARCH) model of this lecture is its starting point

Learning Outcomes

- Explain what a market index must do: be representative, investible and stable
- Compute a Laspeyres index with a divisor and explain why the divisor changes at a rebalancing
- Read an information-criterion curve and explain how CRIX chooses its number of constituents
- Classify a crypto index into one of nine types and read its weighting formula
- Interpret correlation, tracking error and tracking difference between an index, a replica and a fund
- Read compound annual growth rate (CAGR), volatility and drawdown statistics of CRIX, and the VCRIX volatility index

Abbreviations (1)

ADTV average daily traded volume
AIC Akaike information criterion
BGCI Bloomberg Galaxy Crypto Index
BFIC CoinDesk Bitcoin Futures Excess Return Index
BITX Bitwise 10 Large Cap Crypto Index
BTC, ETH, ... coin tickers: Bitcoin, Ethereum, Dogecoin (DOGE), Cardano (ADA), Litecoin (LTC), Bitcoin Cash (BCH), Ethereum Classic (ETC), Monero (XMR), Dash (DASH), Zcash (ZEC)
CAGR compound annual growth rate
CD20 CoinDesk 20 index
CME Chicago Mercantile Exchange
CMC CoinMarketCap
COINTILT Coinbase Size Tilt Index
COVID-19 coronavirus disease 2019
CRIX CRypto IndeX
DACS Digital Asset Classification Standard (CoinDesk)
ECRIX, EFCRIX CRIX variants: first rise / global minimum of the AIC
EODHD EOD (end-of-day) Historical Data, data vendor
ETF exchange-traded fund
ETP exchange-traded product
EVIV Ethereum Volmex Implied Volatility 30 Day Index
EWMA exponentially weighted moving average
FABIZ Faculty of Business Administration in Foreign Languages, ASE
FRED Federal Reserve Economic Data (St. Louis Fed)
FTX crypto exchange (company name), bankrupt in November 2022

Abbreviations (2)

GARCH generalised autoregressive conditional heteroskedasticity
HAC heteroskedasticity- and autocorrelation-consistent
HAR heterogeneous autoregressive (model)
HHI Herfindahl–Hirschman index
IBIT iShares Bitcoin Trust ETF (ticker)
IDA Institute Digital Assets
IQR interquartile range
MBB Master Blockchain for Business
MDVT median daily value traded
MVTT10F MarketVector Token Terminal Fundamental Index
NCI Nasdaq Crypto Index
OLS ordinary least squares
RV realised volatility
S&P Standard & Poor's; **S&P 500**: US stock index; **DAX**: German stock index
SE standard error
SEC US Securities and Exchange Commission
TD, TE tracking difference, tracking error
TMI total market index
USD US dollar; **US** United States
UTC Coordinated Universal Time
VCRIX volatility index of CRIX
VIX Cboe Volatility Index (S&P 500 options)
VWAP volume-weighted average price

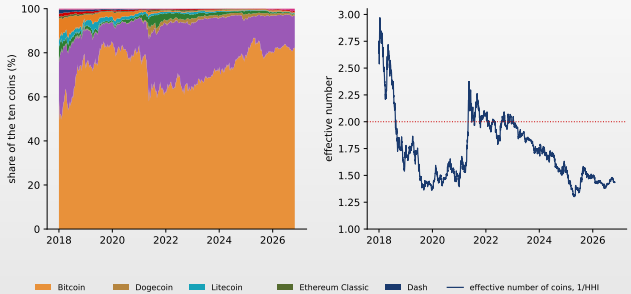
Reading and Tools

- Core reading
  - Trimborn & Härdle (2018), *CRIX an index for cryptocurrencies*, Journal of Empirical Finance 49, 107–122
  - Kim, Trimborn & Härdle (2021), *VCRIX: a volatility index for crypto-currencies*, International Review of Financial Analysis 78, 101915
  - Găman, Będowska-Sójka, Pele & Härdle (2026), *White paper on cryptocurrency indices*, IDA Institute Digital Assets, working paper: types of crypto indices and their methodologies
- Data
  - Royalton CRIX and VCRIX, daily, from the chart data of thecrix.de (CRIX from 6 September 2017, VCRIX from 4 January 2018, to 6 November 2026)
  - prices and volumes from EOD Historical Data (EODHD), to 30 October 2026: ten coins, Bitcoin (BTC), Ethereum (ETH), Dogecoin (DOGE), Cardano (ADA), Litecoin (LTC), Bitcoin Cash (BCH), Ethereum Classic (ETC), Monero (XMR), Dash (DASH), Zcash (ZEC), and the iShares Bitcoin Trust exchange-traded fund (ETF), ticker IBIT
  - coin supplies from the Coin Metrics Community API; effective federal funds rate from FRED: DFF
- Video units on Quantinar (course Blockchain Analytics)
- Further: Trimborn, Li & Härdle (2020); Petukhina et al. (2021)

What an Index Must Do

- **Benchmark**: one number for the performance of a market; funds, traders and researchers compare against it
- Three requirements
  - **representative**: it moves like the whole market, the **total market index** (TMI) of all eligible coins
  - **investible**: few, liquid constituents that a fund can buy and hold at low cost
  - **stable**: rules fixed in advance; constituents do not change every day
- Two features of crypto make this hard
  - **concentration**: one or two coins hold most of the value
  - **entry and exit**: coins appear and vanish; a list of today's coins hides the dead ones
- Trade-off: more constituents represent the market better but cost more to trade; CRIX chooses the number with a statistical rule Trimborn & Härdle (2018)

How Concentrated Is the Market?



- Ten coins with public supply data, market value = price × current supply, 1 January 2018 – 30 October 2026
- Bitcoin: 57.3% of the ten on 1 January 2018, 82.2% on 30 October 2026 (maximum 87.1%, 21 April 2025); above 50% on 99.8% of the days
- Effective number of coins $1/\mathrm{HHI}$, with HHI the Herfindahl–Hirschman index: 2.97 in January 2018, 1.44 today

BA_L3_concentration

Worked Example: the Herfindahl–Hirschman Index

- **HHI** of the value shares $w_i = M_i / \sum_j M_j$:

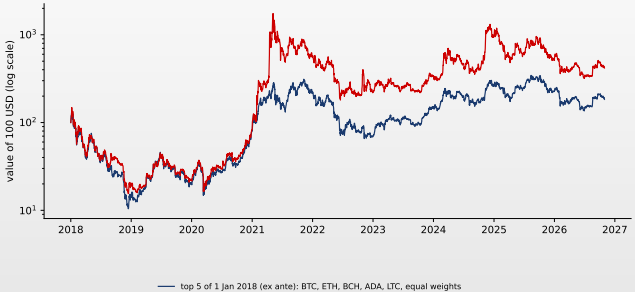
$$\mathrm{HHI} = \sum_{i=1}^{N} w_i^2, \qquad N_{\mathrm{eff}} = 1/\mathrm{HHI}$$

- Ten coins, 30 October 2026: Bitcoin 82.2%, Ethereum 14.5%, the other eight 3.4% together
  - $0.822^2 + 0.145^2 = 0.675 + 0.021 = 0.696$; the eight small coins add almost nothing
  - $N_{\mathrm{eff}} = 1/0.696 = 1.44$: the ten coins behave like 1.4 equal coins
- Consequence for an index
  - a market-value index of these coins is mostly a Bitcoin index
  - a capped index limits the largest constituent, for example below 50% (later in this lecture)

Survivorship Bias

- **Survivorship bias**: a sample built from the assets alive *today* ignores those that failed or shrank Brown et al. (1992)
  - past returns of the survivors look better than what an investor could have earned
- In crypto it is severe: coins are launched, abandoned or delisted every day
- A related error: **look-ahead selection**, choosing constituents with information that was not available at the time
- Remedy built into CRIX: constituents are chosen at each reselection date from the data known on that date Trimborn & Härdle (2018)
- Test on our ten coins
  - ex ante: buy on 1 January 2018 the five largest coins of that day
  - ex post: buy on 1 January 2018 the five largest coins of 30 October 2026

Survivorship Bias: Ex-Ante and Ex-Post Baskets since 2018



- Ex ante (BTC, ETH, BCH, ADA, LTC), equal weights: 100 USD became 184 USD, CAGR 7.1%
- Ex post (BTC, ETH, ZEC, DOGE, XMR): 100 USD became 418 USD, CAGR 17.6%; Zcash ranked 9th of the ten in 2018 and 3rd in 2026
- With market-value weights the gap is smaller (401 against 491 USD), because Bitcoin and Ethereum dominate both baskets

BA_L3_survivorship ▶ Interactive

The Laspeyres Index and the Divisor

- **Laspeyres index**: today's prices of a basket fixed at base time 0

$$\mathrm{CRIX}_t = 1000 \times \frac{\sum_i P_{it}\, Q_{i0}}{\sum_i P_{i0}\, Q_{i0}}$$

  - P_{it}: price of coin i at time t; Q_{i0}: number of coins of i in circulation at time 0
- Market-value form used by CRIX Trimborn & Härdle (2018):

$$\mathrm{CRIX}_t = \frac{\sum_{i=1}^{k} \mathrm{MV}_{it}}{\mathrm{Divisor}_t}, \qquad \mathrm{MV}_{it} = P_{it}\, Q_i, \qquad \mathrm{Divisor}_0 = \frac{\sum_i \mathrm{MV}_{i0}}{1000}$$

  - k: number of constituents; the index starts at 1 000
  - weights are proportional to market value: $w_{it} = \mathrm{MV}_{it} / \sum_j \mathrm{MV}_{jt}$
- Between two rebalancing days the quantities Q_i are fixed; only prices move the index

Why the Divisor Changes

- At a rebalancing day τ the basket changes: a coin enters or leaves, or the quantities Q_i are updated
- The index must not jump because of the change; only prices may move it:

$$\frac{\sum_i \mathrm{MV}_{i\tau}^{\mathsf{old}}}{\mathrm{Divisor}_{\mathsf{old}}} = \mathrm{CRIX}_\tau = \frac{\sum_i \mathrm{MV}_{i\tau}^{\mathsf{new}}}{\mathrm{Divisor}_{\mathsf{new}}} \implies \mathrm{Divisor}_{\mathsf{new}} = \frac{\sum_i \mathrm{MV}_{i\tau}^{\mathsf{new}}}{\mathrm{CRIX}_\tau}$$

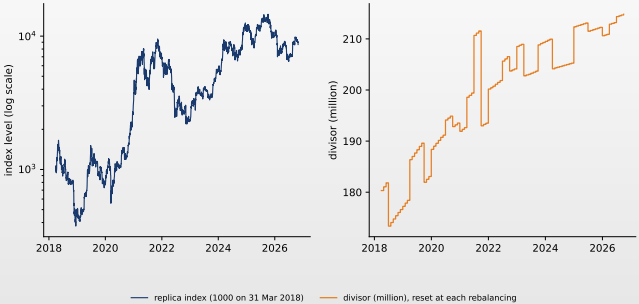
- The same device keeps stock indices such as the S&P 500 (US) or the DAX (Germany) continuous when a company enters or leaves
- CRIX rules thecrix.de
  - **monthly**: the weights are recalculated from current market values (rebalancing)
  - **quarterly**: the constituents are reselected (reconstitution)

Worked Example: a Rebalancing Day of the Replica

Coin	price, 30 Sep 2026 (USD)	supply (million)	market value (billion USD)	weight (%)	price, 30 Oct 2026 (USD)
Bitcoin	83 553.85	20.09	1 678.74	82.07	77 198.21
Ethereum	2 683.68	122.09	327.65	16.02	2 241.05
Zcash	1 436.40	16.90	24.27	1.19	1 268.64
Dogecoin	0.0945	156 136	14.75	0.72	0.0759

- Our replica (later in this lecture) on its quarterly reselection day, 30 September 2026: index level $\mathrm{CRIX}_\tau = 9\,521.15$
- Step 1: new basket value $\sum_i \mathrm{MV}_{i\tau} = 2\,045.41$ billion USD
- Step 2: $\mathrm{Divisor}_{\mathbf{new}} = 2\,045.41 \times 10^{\mathbf{9}} / 9\,521.15 = 214.83$ million (before: 214.67 million)
- Step 3, 30 October 2026: same quantities, new prices, $\sum_i P_{it} Q_i = 1\,857.94$ billion USD
  - $\mathrm{CRIX}_t = 1\,857.94 \times 10^{\mathbf{9}} / (214.83 \times 10^{\mathbf{6}}) = 8\,648.5$, a fall of 9.2% in one month

The Divisor through Time



- Left: the replica index, 1 000 on 31 March 2018; right: its divisor, recomputed at each of the 103 monthly rebalancing days
- The divisor moves in steps (new coins, new supplies); the index level is unchanged on each of these days
- Rising supply (new Bitcoin, Ethereum and Dogecoin issued) raises $\sum_i \mathrm{MV}_{i\tau}$ and with it the divisor

BA_L3_divisor

Which Coins Are Eligible?

- **Liquidity rule** thecrix.de: coin i is eligible if

$$\mathrm{ADTV}_i \geq \mathrm{ADTV}_{0.25}$$

  - ADTV_i: average daily traded volume (USD) of coin i in the last period
  - $\mathrm{ADTV}_{0.25}$: the 25th percentile of the ADTVs of all coins: the least traded quarter is excluded
- Stablecoins are excluded: their price is pegged, not formed by the market Royalton Partners: CRIX
- Eligible coins are ranked by market value; the candidate indices take the first k of the ranking
- In our replica (90-day window): Monero failed the liquidity rule in 33 of 35 quarters, Dash in 24, Zcash in 22

How Many Constituents? The Information Criterion

- Candidate index $\mathrm{CRIX}(k, \beta)$: the largest coin with weight 1, the next $k-1$ with free weights β Trimborn & Härdle (2018); its tracking residual:

$$\hat{\varepsilon}_t(k, \beta) = r_t^{\mathrm{TMI}} - r_t^{\mathrm{CRIX}(k,\beta)}, \qquad r_t = \ln I_t - \ln I_{t-\mathbf{1}}$$

- Density of the residuals of the one-coin index, by a Gaussian kernel with leave-one-out:

$$\hat{f}_{-t}(x) = \frac{1}{(n-1)h} \sum_{s \neq t} \varphi\Big(\frac{x - \hat{\varepsilon}_s(1,1)}{h}\Big), \qquad h = 0.9 \, \min\left(\hat{\sigma}, \widehat{\mathrm{IQR}}/1.349\right) n^{-\mathbf{1/5}}$$

  - φ: standard Normal density; IQR: interquartile range; $\hat{\sigma}$: standard deviation

- Akaike information criterion (AIC) Akaike (1974) of the candidate with k constituents:

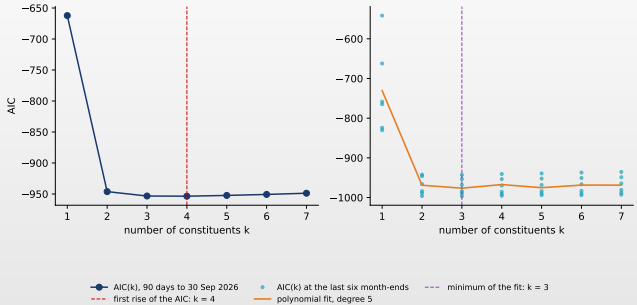
$$\mathrm{AIC}(k) = -2 \sum_{t=\mathbf{1}}^{n} \ln \hat{f}\left(\hat{\varepsilon}_t(k, \hat{\beta})\right) + 2\,(k-1)$$

  - fit: more coins push the residuals towards 0, where the density is highest; penalty: 2 per added coin
  - ECRIX rule: stop at the first k where the AIC rises; EFCRIX: the global minimum

Smoothing the Decision over Six Months

- A single-day AIC curve is noisy: a small change of prices can move its minimum from one k to another
  - an index whose constituents change for such reasons is costly for a fund that follows it
- Smoothed rule, used in our replica
  - compute the AIC curve at each of the last six month-ends, each on its own 90-day window
  - pool the six curves and fit polynomials in k of degree 1 to $k_{\max} - 2$ through them
  - keep the degree with the lowest AIC of the fit; the minimum of the fitted polynomial gives k
- The price: the rule reacts later when the market really changes

The Decision Curve on Our Data



- Left: $\mathrm{AIC}(k)$ on the 90 days to 30 September 2026; seven coins pass the liquidity rule (ETC, XMR, DASH excluded)
- From $k = 1$ to $k = 2$ the AIC falls by 284: adding Ethereum is decisive; after $k = 3$ the curve is almost flat
- Right: our smoothed rule, six month-end curves and a polynomial of degree 5; minimum at $k = 3$

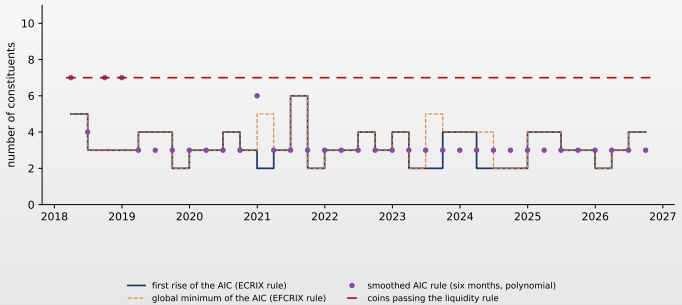
BA_L3_aic_curve ▶ Interactive

Worked Example: Reading the AIC Values

k	1	2	3	4	5	6	7
$\mathrm{AIC}(k)$	-662.2	-946.2	-953.3	-953.6	-952.3	-950.7	-948.8

- 90 days to 30 September 2026; eligible coins by market value: BTC, ETH, ZEC, DOGE, ADA, BCH, LTC
- Step 1: the AIC falls until $k = 4$ (-953.6) and rises at $k = 5$ (-952.3): ECRIX chooses $k = 4$
- Step 2: the global minimum is also at $k = 4$: EFCRIX agrees
- Step 3: the gain from the 4th coin is $953.6 - 953.3 = 0.3$ AIC points, against a gain of 284 from the 2nd
  - a flat curve: the choice between 3 and 4 is fragile; this is the reason for the smoothing

The Number of Constituents, Quarter by Quarter



- 35 quarterly reselections, 31 March 2018 – 30 September 2026; on average 7 of the 10 coins pass the liquidity rule
- First rise of the AIC: $k = 2$ in 9 quarters, 3 in 13, 4 in 11, 5 or 6 once each; mean 3.2
- The smoothed rule chooses $k = 3$ in 30 of 35 quarters: the smoothing makes the index more stable

BA_L3_k_choice

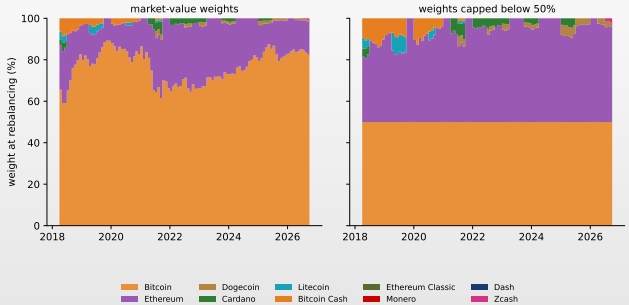
Capped Weights

- A capped market-value index:

$$\mathrm{CRIX}(k)_t = \frac{\sum_{i=1}^{k} \mathrm{MV}_{it}\,\mathrm{AW}_{it}}{\mathrm{Divisor}}, \qquad \mathrm{AW}_{it} = \frac{\mathrm{CW}_{it}}{W_{it}}, \qquad W_{it} = \frac{\mathrm{MV}_{it}}{\sum_j \mathrm{MV}_{jt}}$$

  - W_{it}: market-value weight; CW_{it}: capped weight, so that no coin has an influence of 50% or more
  - AW_{it}: the adjustment factor applied to the market value
- Capping algorithm (as in other capped indices): set the weight above the cap to the cap; share the excess among the other constituents in proportion to their weights; repeat
- Worked example, replica on 30 September 2026
  - market-value weights: BTC 82.1%, ETH 16.0%, ZEC 1.2%, DOGE 0.7%
  - capped below 50%: BTC 49.99%, ETH 44.7%, ZEC 3.3%, DOGE 2.0%; the excess of 32.1 points goes mostly to Ethereum

Market-Value Weights against Capped Weights



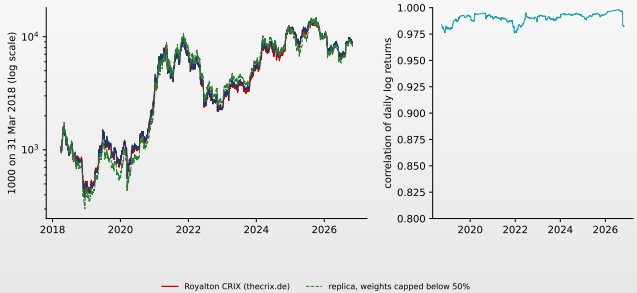
- Weights of the replica at each monthly rebalancing, April 2018 – September 2026
- Market-value weight of Bitcoin: from 59.1% to 89.3%, mean 76.9%; the 50% cap would bind at every rebalancing in this universe
- The capped index is a different product: much more Ethereum, less Bitcoin

BA_L3_replica_weights

Replication Design

- Universe: ten coins whose circulating supply is public (Coin Metrics Community API): BTC, ETH, DOGE, ADA, LTC, BCH, ETC, XMR, DASH, ZEC
  - prices and volumes in USD: EODHD daily closes; market value = price $\times$ current supply
  - not included: coins without a public supply history in this source (e.g. BNB, SOL) or whose reported supply includes issuer holdings (XRP)
- Rules, as in Trimborn & Härdle (2018) and thecrix.de
  - quarterly reselection (31 March 2018 – 30 September 2026): liquidity rule on the last 90 days, ranking by market value
  - number of constituents: first rise of the AIC on the last 90 days (ECRIX)
  - monthly rebalancing with new supplies; divisor; start at 1 000 on 31 March 2018
- Benchmarks: the TMI of the eligible coins, Bitcoin alone, a version with weights capped below 50%
- Comparison with the published CRIX on common days, 1 April 2018 – 30 October 2026, $n = 3\,105$ daily log returns

The Replica against the Real CRIX



- Both start at 1 000 on 31 March 2018; on 30 October 2026: CRIX 8 380, replica 8 648, capped replica 8 153
- 180-day correlation of daily log returns between 0.976 (December 2021) and 0.998
- Ten coins and a handful of constituents reproduce most of the published index

BA_L3_replica_vs_crix ▶ Interactive

How Close Is the Replica?

Series compared with the published CRIX	correlation	tracking error (% a year)	beta
Replica (ECRIX rule, market-value weights)	0.990	9.3	1.00
TMI of the eligible coins	0.990	9.1	0.99
Replica, weights capped below 50%	0.979	14.5	0.91
Bitcoin alone	0.971	15.6	1.02

- Daily log returns, 1 April 2018 – 30 October 2026, $n = 3\,105$; beta: slope of CRIX returns on the series' returns
- Interpretation
  - the replica tracks its own TMI closely: correlation 0.9995, tracking error 2.2% a year
  - the remaining gap to CRIX comes from the universe (coins we do not have), not from the rule
  - on these data the uncapped replica is closer to the published CRIX than the capped one
  - a basket of a few coins tracks the market better than Bitcoin alone

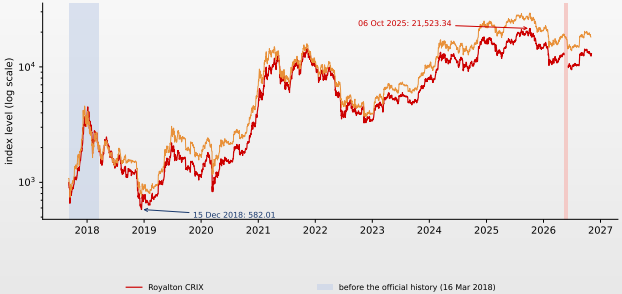
Tracking Error and Tracking Difference

- **Tracking error** (TE) and **tracking difference** (TD) of a portfolio (replica or fund) with daily log returns r_t^P against its benchmark r_t^B:

$$\mathrm{TE} = \sqrt{A}\,\mathrm{sd}(r_t^P - r_t^B), \qquad \mathrm{TD} = \frac{A}{n}\sum_{t=1}^{n}(r_t^P - r_t^B)$$

  - TE: how much the portfolio wanders around the benchmark; TD: by how much it lags or leads on average
  - A: returns per year (365 for coins, 252 for exchange-traded funds)
- Worked example: replica against CRIX, TE $= 9.3\%$ a year
  - daily: $9.3/\sqrt{365} = 0.49$ percentage points
  - a typical day: if CRIX moves by 3%, the replica moves by about 2.5% to 3.5%
- A good index fund has a small TE (it follows the index) and a TD close to minus its fees

CRIX, 2017–2026



- Daily values from thecrix.de, 6 September 2017 – 6 November 2026; no values on 22 March 2026 and from 14 May to 7 June 2026
- Low 582.01 on 15 December 2018; record 21 523.34 on 6 October 2025; last value 12 250.34 (6 November 2026), 43.1% below the record
- Before 16 March 2018 the series comes from a different data source and is shifted by one day against Bitcoin; statistics below start on 16 March 2018

BA_L3_crix_path ▶ Interactive

CRIX and Bitcoin in Numbers

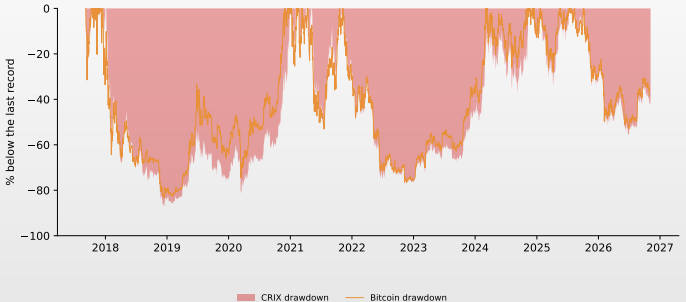
16 Mar 2018 – 31 Oct 2026	CAGR (%)	ann. vol. (%)	max. drawdown (%)	skewness	exc. kurtosis	worst day (%)
CRIX	24.0	65.0	−77.4	−1.09	14.4	−46.5
Bitcoin	29.4	61.7	−76.6	–	–	–

- Daily log returns, $A = 365$; CRIX: $n = 3\,121$ returns between consecutive days; worst day 12 March 2020, best day 2 April 2019 (+16.8%)
- Regression of CRIX returns on Bitcoin returns ($n = 3\,122$ common days): beta 1.020 (standard error, SE, 0.005), $R^2 = 0.94$, correlation 0.971
- Interpretation
  - CRIX is slightly *more* volatile than Bitcoin: the other coins add volatility, not diversification
  - over this period CRIX grew less than Bitcoin (24.0% against 29.4% a year): Bitcoin's share of the market rose

Worked Example: CAGR and the Last Drawdown

- **CAGR**: $(I_T/I_0)^{1/T} - 1$ with T in years
  - CRIX: $I_0 = 1\,960.12$ (16 March 2018), $I_T = 12\,512.61$ (31 October 2026), $T = 8.63$
  - $(12\,512.61/1\,960.12)^{1/8.63} - 1 = 6.384^{0.116} - 1 = 24.0\%$ a year
  - Bitcoin, same dates: $(77\,214/8\,338)^{1/8.63} - 1 = 29.4\%$ a year
- **Drawdown** $D_t = I_t / \max_{s \le t} I_s - 1$
  - record 21 523.34 on 6 October 2025; trough 9 439.22 on 30 June 2026: $D = 9\,439.22/21\,523.34 - 1 = -56.1\%$
  - gain needed to recover: $21\,523.34/9\,439.22 - 1 = 128\%$
  - 6 November 2026: 12 250.34, still 43.1% below the record

Drawdowns of CRIX and Bitcoin



- The two drawdown curves almost coincide; CRIX falls a little deeper in each crash
- Since September 2017 CRIX was more than 50% below its record on 48.8% of the days, Bitcoin on 39.9%

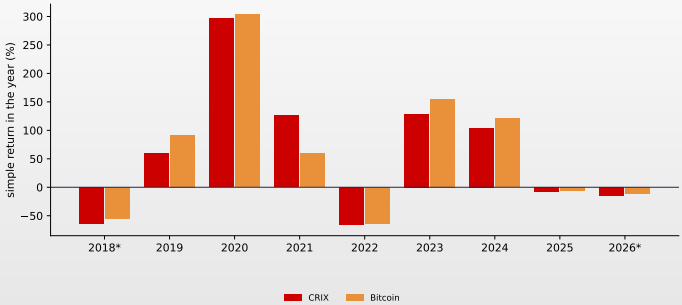
BA_L3_crix_drawdown ▶ Interactive

Episodes: Crypto Winters and Shocks

Peak	level	trough	depth (%)	new record	days peak to record
5 Jan 2018	4 510.28	15 Dec 2018 (582.01)	−87.1	31 Dec 2020	1 091
9 May 2021	12 644.70	20 Jul 2021 (5 994.23)	−52.6	19 Oct 2021	163
8 Nov 2021	14 484.51	9 Nov 2022 (3 268.34)	−77.4	11 Nov 2024	1 099
6 Oct 2025	21 523.34	30 Jun 2026 (9 439.22)	−56.1	not yet	–

- Falls of CRIX deeper than 50%; the 2018 peak lies before the official history
- Three shocks, cumulative log returns (same windows as Lecture 2)
  - COVID-19, 19 February – 23 March 2020: CRIX −45.0, Bitcoin −40.6
  - Terra/Luna, 5 – 12 May 2022: CRIX −31.0, Bitcoin −23.0
  - FTX, 5 – 21 November 2022: CRIX −33.8, Bitcoin −29.9
- In crypto-specific shocks the broad index loses more than Bitcoin: smaller coins fall harder

Year by Year: CRIX against Bitcoin



- Simple returns per calendar year; *2018 from 16 March, 2026 to 31 October
- CRIX beat Bitcoin only in 2021 (+125.7% against +59.7%), the year of the large altcoin rally
- In falling years (2018, 2022, 2025, 2026) CRIX lost more than Bitcoin

BA_L3_crix_years

Motivation for a Volatility Index

- Equity markets have a "fear gauge": the Cboe Volatility Index (VIX), the 30-day volatility implied by S&P 500 option prices Whaley (2000)
- For the crypto market as a whole there was no liquid option market on an index, hence no implied volatility
- Kim, Trimborn & Härdle (2021): VCRIX, a forecast of CRIX volatility that plays the role of implied volatility
  - proxy for implied volatility: the realised volatility of CRIX over a 30-day window
  - forecasting model chosen by out-of-sample accuracy among GARCH models, HAR and a neural network; HAR performed best
- Uses
  - a daily risk thermometer for the whole market
  - an input for risk limits, option pricing and volatility-timing strategies

VCRIX: Construction

- Realised volatility (RV) of CRIX over the last $w = 30$ days, annualised, with r_t the daily log return in %:

$$\mathrm{RV}_t^{(w)} = \sqrt{\frac{365}{w} \sum_{j=0}^{w-1} r_{t-j}^2}$$

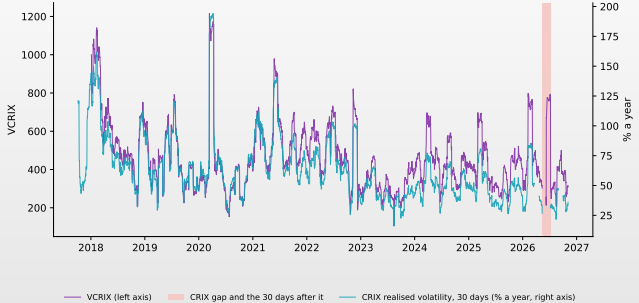
- Heterogeneous autoregressive (HAR) model Corsi (2009): daily, weekly and monthly components forecast the next value

$$\mathrm{RV}_{t+1} = c + \beta_d \, \mathrm{RV}_t^{(1)} + \beta_w \, \mathrm{RV}_t^{(7)} + \beta_m \, \mathrm{RV}_t^{(30)} + u_{t+1}$$

  - three horizons of traders: day traders, weekly and monthly investors
  - estimated by ordinary least squares (OLS); the forecast $\widehat{\mathrm{RV}}_{t+1}$ is the volatility expected for tomorrow

- The index: $\mathrm{VCRIX}_t = \widehat{\mathrm{RV}}_t / \mathrm{Divisor}$, with the divisor set so that VCRIX $= 1\,000$ on the start date (4 January 2018 in the published series)

VCRIX, 2018–2026



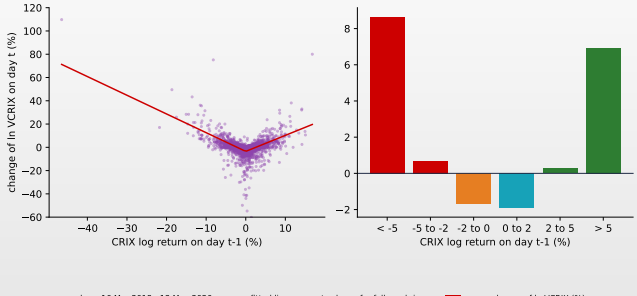
- VCRIX (thecrix.de) and the 30-day realised volatility of CRIX: correlation 0.943 with the realised volatility of the previous day ($n = 3\,020$)
- Mean 458, median 427; maximum 1 215.57 on 13 March 2020, minimum 105.09 on 14 August 2023; 342.13 on 6 November 2026
- VCRIX $\approx 74.4 + 6.11\,\mathrm{RV}_{t-\mathbf{1}}^{(\mathbf{30})}$ ($R^{\mathbf{2}} = 0.89$): 50% realised volatility corresponds to a VCRIX of about 380

BA_L3_vcrix_path ▶ Interactive

Worked Example: the 30-Day Window

- 12 March 2020: CRIX log return -46.5%
  - VCRIX 405.66 on 12 March, 1 215.57 on 13 March: the crash enters the window one day later
  - one day enters the 30-day sum: $46.5^2/30 = 72$ added to the mean square, i.e. $\sqrt{365 \times 72} = 162\%$ of annual volatility from one day
- 30 days later: VCRIX 1 159.78 on 11 April, 542.82 on 12 April 2020
  - the crash leaves the window; nothing happened in the market that day
- Data gap of 14 May – 7 June 2026: VCRIX jumps from 213.04 to 673.24 on 10 June and falls from 793.12 to 283.93 on 9 July
  - a return over a 25-day gap inside a 30-day window: a data artefact; this window is excluded from our statistics
- Lesson: a rolling window gives a "ghost" effect; a GARCH model lets old shocks fade gradually

Volatility after Falls and Rises



- Change of ln VCRIX on day t against the CRIX return on day $t-1$, 16 March 2018 – 13 May 2026, $n = 2\,978$; correlation -0.132 (same day: 0.007)
- Regression with heteroskedasticity- and autocorrelation-consistent (HAC) SE: slope -1.61 (SE 0.14) after falls, $+1.38$ (SE 0.12) after rises; equal size not rejected (p 0.10)
- Large moves in *either* direction raise VCRIX: after a fall beyond 5% by 8.6% on average, after a rise beyond 5% by 6.9%

BA_L3_vcrix_leverage ▶ Interactive

A GARCH(1,1) Volatility for CRIX

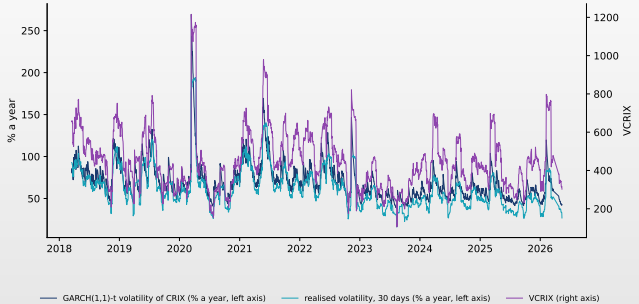
- GARCH(1,1) Bollerslev (1986) with Student-t errors for the daily log returns r_t (%) of CRIX:

$$r_t = \mu + \sigma_t z_t, \quad z_t \sim t_\nu \text{ (unit variance)}, \qquad \sigma_t^2 = \omega + \alpha\,(r_{t-1} - \mu)^2 + \beta\,\sigma_{t-1}^2$$

	μ	ω	α	β	ν
estimate	0.128	0.254	0.084	0.914	3.14
SE (robust)	0.042	0.107	0.020	0.019	0.20

- 16 March 2018 – 13 May 2026, $n = 2\,979$ returns (the two returns around the missing 22 March 2026 skipped)
- Persistence $\alpha + \beta = 0.998$: a shock to variance halves in $\ln 0.5 / \ln 0.998 \approx 370$ days
- $\hat{\nu} = 3.1$: very heavy tails, as in Lecture 2
- Annualised GARCH volatility $\sqrt{365}\,\hat{\sigma}_{t+1}$: mean 69.8%, maximum 269% on 13 March 2020
- Lecture 4 develops GARCH, value at risk and expected shortfall

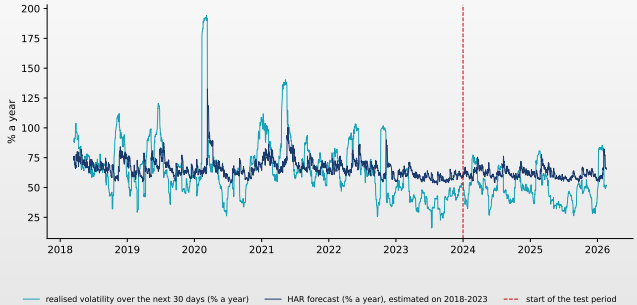
VCRIX against GARCH and Realised Volatility



- Correlations, $n = 2\,949$ days: VCRIX with the 30-day realised volatility 0.916, with the one-day-ahead GARCH forecast 0.844; GARCH with realised volatility 0.908
- The three measures agree on the regimes (2018, March 2020, May 2021, November 2022); VCRIX follows the 30-day window, GARCH reacts faster and fades smoothly

BA_L3_vcrix_garch ▶ Interactive

Our HAR Forecast of the Next Month's Volatility



- Target: realised volatility of CRIX over the next 30 days; training 16 March 2018 – 31 December 2023 ($n = 2\,117$), test 2024 – February 2026 ($n = 781$)
- Estimates (HAC SE): $c = 48.0$ (6.5), $\beta_d = 0.021$ (0.007), $\beta_w = 0.112$ (0.042), $\beta_m = 0.153$ (0.089); in-sample $R^2 = 0.09$
- Test period: root mean squared error 18.4 points, against 18.6 for "next month = last month"; the forecast correlates 0.73 with VCRIX
- A month ahead, crypto volatility is hard to forecast: the forecast stays close to the long-run mean

BA_L3_vcrix_har

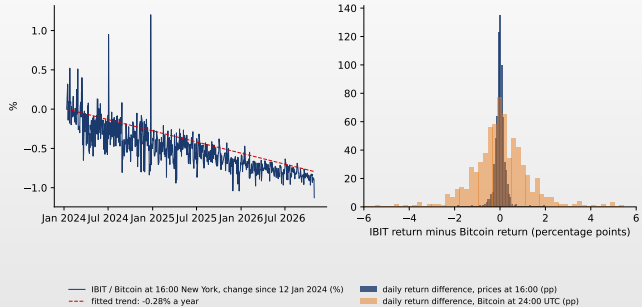
From an Academic Index to an Investible Index

- 2015–2016: CRIX discussion papers, Humboldt-Universität zu Berlin
- 2018: Trimborn & Härdle (2018) publish the method; official history of the index starts on 16 March 2018
  Royalton Partners: CRIX
- 2021: Royalton Partners AG acquires the rights to the index
- From November 2021: the index is run commercially as the Royalton CRIX Crypto Index
  - Royalton Partners: CRIX: calculated by Royalton Partners from November 2021; thecrix.de: calculated by S&P Global since November 2021
  - target: track the TMI "with as few as possible but more than 5 liquid cryptocurrencies"; stablecoins excluded
    Royalton Partners: CRIX
- 2021: Kim, Trimborn & Härdle (2021) add VCRIX, the volatility index

From an Index to a Fund

- An index is a number; a fund holds the coins
  - the index provider publishes constituents and weights
  - the fund issues shares, takes investors' money and buys the constituents in the index weights
- **Exchange-traded product** (ETP): a fund or note whose shares trade on a stock exchange
  - 10 January 2024: the US Securities and Exchange Commission (SEC) approved spot Bitcoin ETFs in the US SEC (2024)
  - a spot ETF holds the coins themselves, not futures
- Why a fund deviates from its index
  - fees (a steady drag: tracking difference)
  - cash, trading costs, timing of rebalancing, different price sources (noise: tracking error)
  - small constituents may be illiquid: hence the liquidity rule and a small number of constituents

A Spot Bitcoin Fund against Bitcoin



- IBIT (EODHD, adjusted close) against Bitcoin at 16:00 New York time (EODHD hourly), 12 January 2024 – 30 October 2026, $n = 697$ trading days
- Left: IBIT per Bitcoin fell by 1.1%; trend −0.28% a year, close to the sponsor fee of 0.25% iShares: IBIT
- Right: daily return differences; TE 3.5% a year with synchronous prices, 20.6% with Bitcoin's close at midnight, Coordinated Universal Time (UTC)

BA_L3_etf_tracking ▶ Interactive

Worked Example: Tracking Error and the Benchmark Closing Time

- IBIT trades 9:30–16:00 New York time on weekdays; Bitcoin trades 24 hours a day, 7 days a week
- Same days, two Bitcoin prices
  - at 16:00 New York: sd of daily differences 0.223 percentage points, $\mathrm{TE} = 0.223 \times \sqrt{252} = 3.5\%$; correlation 0.998
  - at 24:00 UTC (EODHD daily close): sd 1.299 points, $\mathrm{TE} = 1.299 \times \sqrt{252} = 20.6\%$; correlation 0.907
- The fund did not change; only the clock of the benchmark did
- Tracking difference: -0.28% a year, IBIT $+73.6\%$ against Bitcoin $+75.6\%$ over the period
- Lesson for index products: compare a fund with its index at the time the fund is priced, then read TE and TD separately

Nine Types of Crypto Indices

Type	what it tracks	example
Multi-asset	several coins in one number, usually by market value	BITX; CRIX
Sector	one segment: DeFi, smart-contract platforms, Metaverse	CoinDesk Metaverse Select
Price	the spot price of one coin, from several exchanges	MarketVector Polygon VWAP Close
Futures	futures contracts, rolled from one maturity to the next	CoinDesk BFIC
Volatility	expected or realised variability: a "fear gauge"	EVIV; VCRIX
Yield	income from cash, lending or staking, with crypto exposure	LIXX Risk-Adjusted Bitcoin
Smart beta	rule-based weights other than market value	COINTILT
Fundamentals	on-chain activity: fees, active users	MVTT10F
Sentiment	market and behavioural signals in one score	CMC Fear and Greed

- Taxonomy and examples from Găman, Będowska-Sójka, Pele & Härdle (2026); DeFi: decentralised finance
- Most of them share one backbone, the Laspeyres index with a divisor; they differ in universe, weights, data and rules

Multi-Asset: the Bitwise 10 (BITX)

- The ten largest eligible coins by free-float market value Bitwise (2025), Găman, Będowska-Sójka, Pele & Härdle (2026)

$$I_t = \frac{1}{D_t} \sum_{i \in \mathcal{I}_t} P_{it}\, Q_{it}^{\mathbf{ff}}$$

  - $Q_{it}^{\mathbf{ff}}$: free-float supply (coins that can trade); D_t: divisor; $I = 964$ on 1 January 2017, the price of Bitcoin that day
  - weights rescaled so that at least 87.5% go to coins with an SEC-approved single-asset ETP
- Eligible trading venues
  - regulated, no capital controls, secure interfaces, at least 0.1% of global crypto volume
- Eligible coins
  - not pegged, listed on several approved venues, no critical security flaws
  - 30-day volume above 1% of the free-float market value; low risk of being a security under US law
- Monthly reconstitution; a committee and an advisory board review the rules
- CRIX differs on one point: the number of constituents is chosen by the data, not fixed at ten

Sector and Price Indices

- **Sector**: CoinDesk Metaverse Select CoinDesk (2024)
  - coins of the Metaverse group of the Digital Asset Classification Standard (DACS); no meme coins; custody and US access required
  - ranked by size; liquidity by the median daily value traded (MDVT) over 30 days; 5 to 10 coins; minimum weight 1%

$$\text{Index}_t = \frac{\sum_{i=\mathbf{1}}^{N} P_{it}\, S_i\, \text{WAF}_i}{\text{Divisor}}$$

  - S_i: circulating supply at the weighting date; WAF_i: weighting adjustment factor; quarterly reconstitution
- **Price**: MarketVector Polygon VWAP Close MarketVector (2025a)
  - one coin, Polygon (MATIC), weight 100%; close = one-hour volume-weighted average price (VWAP), 15:00–16:00 Central European Time
  - base value 100 on 30 September 2020; after a hard fork the new coins may enter for a while, weighted by market value

$$\text{Index}_t = \frac{\sum_{i=\mathbf{1}}^{n} p_i\, q_i\, cf_i\, fx_i}{D}, \qquad \text{VWAP} = \frac{\sum_j p_j v_j}{\sum_j v_j}$$

  - cf_i: cap factor (usually 1); fx_i: exchange rate to USD; p_j, v_j: price and volume of trade j in the hour

Futures: Rolling CME Contracts

- CoinDesk Bitcoin Futures Excess Return Index (BFIC) CoinDesk (2025), Găman, Będowska-Sójka, Pele & Härdle (2026)
  - Bitcoin futures of the Chicago Mercantile Exchange (CME); launched 7 February 2024, base 14 690.00 on 27 December 2017
  - holds the front contract; before it expires, exposure moves to the next contract (the roll)

$$I_t = U_{L,t}\, P_{L,t} + U_{N,t}\, P_{N,t}, \qquad U_{L,t} = U_{L,\mathbf{start}} - r, \quad U_{N,t} = r$$

  - U: units of the lead (L) and next (N) contract; P: CME settlement price at 16:00 New York time
  - the roll starts on the Friday before the last trade date and ends on the Wednesday: a quarter of the position per day over four days
- No market-value weights: exposure is set in contract units
- The return has two parts: the change of the futures price and the roll yield
  - if the next contract is dearer than the lead (contango), each roll buys fewer units: a drag against spot Bitcoin

Volatility: Implied Volatility from Options

- Ethereum Volmex Implied Volatility 30 Day Index (EVIV) Volmex, Găman, Będowska-Sójka, Pele & Härdle (2026): the 30-day expected volatility from Ether options
- Variance of each maturity from out-of-the-money calls and puts, as for the VIX Whaley (2000):

$$\sigma_i^2 = \frac{1}{T_i}\Big(2\sum_j \frac{\Delta K_j}{K_j^2}\, e^{r_i T_i}\, V(K_j) - \Big(\frac{F_i}{K_{i,\mathbf{ATM}}} - 1\Big)^2\Big), \qquad i \in \{\text{near}, \text{next}\}$$

  - $V(K_j)$: price of the option with strike K_j; F_i: forward price; $K_{i,\mathbf{ATM}}$: at-the-money strike
- Interpolation to 30 days, $\sigma^2_{\mathbf{raw}} = \omega_{\mathbf{near}}\sigma^2_{\mathbf{near}} + \omega_{\mathbf{next}}\sigma^2_{\mathbf{next}}$, then smoothing by an exponentially weighted moving average (EWMA):

$$\sigma^2_{\mathbf{smooth},t} = \lambda\, \sigma^2_{\mathbf{smooth},t-\mathbf{1}} + (1-\lambda)\, \sigma^2_{\mathbf{raw},t}, \qquad \lambda = e^{-\ln \mathbf{2}/\tau}, \qquad \mathrm{EVIV}_t = 100\, \sigma_{\mathbf{smooth},t}$$

  - τ: half-life; an old value loses half of its weight after τ periods
- VCRIX (this lecture) has no options to use: it forecasts the volatility of CRIX from its own past with a HAR model

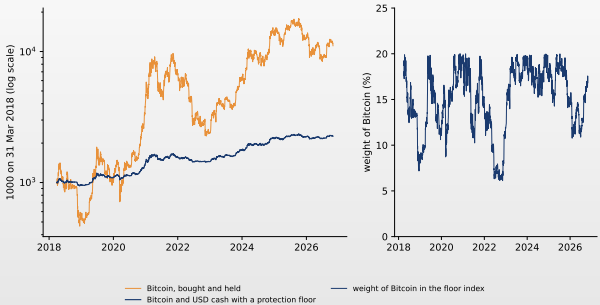
Yield: Bitcoin and Cash with a Protection Floor

- LIXX Risk-Adjusted Bitcoin & USD Overnight Interest Index LIXX (2023), Găman, Będowska-Sójka, Pele & Härdle (2026)
  - two parts: Bitcoin, and USD cash that earns the overnight rate; start: 19% Bitcoin, 81% cash
- Protection floor, checked every day:

$$\text{Floor}_t = (\text{lock-in} + \text{buffer}) \times \max_{s \in [t - \mathbf{1y},\, t]} I_s, \qquad \text{lock-in} = 80\%,\ \text{buffer} = 1\%$$

  - rebalance when the gap between the floor and the cash exceeds the buffer: sell Bitcoin after falls, buy after rises
  - a fee of 0.3% of the traded amount
- Index value: $I_t = \sum_i W_{it}\, P_{it} - A_t$
  - W_{it}: units of Bitcoin and of cash; A_t: adjustment for forks or changes of method
- Our reading of the rule, on real data: cash = floor, so the start is 81% cash, as published; cash earns the effective federal funds rate FRED: DFF

The Protection Floor on Real Data



- 31 March 2018 – 30 October 2026: floor index CAGR 9.9%, volatility 9.1%, maximum drawdown −13.9%; Bitcoin: 32.3%, 61.6%, −76.6%
- Weight of Bitcoin between 6.1% and 20.0%; 66 rebalancings in 8.6 years

BA_L3_floor

Smart Beta and Fundamentals

- **Smart beta**: Coinbase Size Tilt Index (COINTILT) MarketVector (2025b)

$$w_i = \frac{\sqrt{\mathrm{MV}_i}}{\sum_{j=\mathbf{1}}^{n} \sqrt{\mathrm{MV}_j}}$$

  - the square root tilts the index towards smaller coins; no cap
  - eligibility: at least 0.25% of the market value of the MarketVector Digital Assets 100, 180 days of trading, 25 million USD a day of volume, custody by Coinbase-approved custodians

- **Fundamentals**: MarketVector Token Terminal Fundamental Index (MVTT10F) MarketVector (2024)

$$w_i = \sum_{k=\mathbf{1}}^{m} a_k \, w_{k,i}, \qquad w_{k,i} = \frac{k_i}{\sum_{j=\mathbf{1}}^{n} k_j}$$

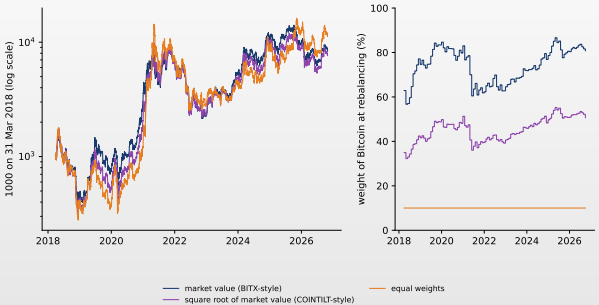
  - two on-chain factors k: fees, $a = 80\%$, and active users, $a = 20\%$; each the monthly mean of the square roots of the daily values
  - ten coins: the top seven by the factor ranking, then three from ranks 8 to 13; meme and privacy coins excluded

- Both use the same Laspeyres calculation as BITX and CRIX; only the weights change

Worked Example: Square-Root Weights

- Ten coins, 30 September 2026, market values in billion USD: BTC 1 678.7, ETH 327.7, the other eight 81.7 together
- Market-value weights: BTC $1\,678.7/2\,088.1 = 80.8\%$, ETH 15.8%
- Square roots: $\sqrt{1\,678.7} = 41.0$, $\sqrt{327.7} = 18.1$; sum over the ten coins 80.9
  - BTC $41.0/80.9 = 50.7\%$, ETH $18.1/80.9 = 22.4\%$, the other eight 26.9% together
- Effective number of coins, $1/\mathrm{HHI}$: 1.48 with market-value weights, 3.15 with square-root weights, 10 with equal weights
- The square root does not change the ranking; it compresses the distance between large and small coins

Three Weightings of the Same Ten Coins



- Monthly rebalancing, 31 March 2018 – 30 October 2026; CAGR / volatility / maximum drawdown: market value 28.0% / 64.6% / −77.9%; square root 26.7% / 69.7% / −83.4%; equal 32.8% / 80.0% / −84.5%
- Correlation of daily returns with the market-value index: 0.980 (square root), 0.880 (equal weights)

BA_L3_weighting

Sentiment: the Fear and Greed Index

- CMC Crypto Fear and Greed Index CoinMarketCap (2025), Găman, Będowska-Sójka, Pele & Härdle (2026): one daily score from 0 (extreme fear) to 100 (extreme greed)

$$\text{Score}_t = \sum_{k=1}^{m} w_k \, s_{kt}, \qquad s_{kt} \in [0, 100]$$

- Components and weights w_k
  - volatility (implied volatility of Bitcoin and Ether options) 25%; market momentum and volume 25%
  - social media 15%; surveys 15%; Bitcoin dominance 10%; search trends 10%
- Bands: 0–24 extreme fear, 25–49 fear, 50 neutral, 51–74 greed, 75–100 extreme greed
- Not an investible index: no basket, no divisor; a summary of signals, and the choice of weights is the provider's judgement

Broad Market Index Families

Index	constituents and weights	source
Royalton CRIX	number chosen by an information criterion; market-value weights; monthly rebalancing, quarterly re-election	thecrix.de
S&P Cryptocurrency Broad Digital Market	over 200 coins (S&P Dow Jones Indices, since 2021)	S&P Dow Jones Indices: Cryptocurrency Indices
Bloomberg Galaxy Crypto Index (BGCI)	largest eligible coins; weight cap 40%, floor 1%; monthly	Bloomberg (2020)
CoinDesk 20 (CD20)	largest 20 coins, no stablecoins or memecoins; cap 30% for the largest, 20% for the others; quarterly	CoinDesk Indices (2025)
Nasdaq Crypto Index (NCI)	benchmark of a significant part of the digital-asset market	Nasdaq: NCI

- Descriptive only, from the providers' documents; methodologies change, check the current version
- The main design choices: fixed or data-driven number of constituents, cap or no cap, frequency of review

Summary

- The crypto market is concentrated: Bitcoin holds 82% of our ten coins, an effective number of 1.4 coins
- A list of today's winners overstates past returns: 17.6% a year against 7.1% for the coins chosen ex ante
- CRIX: a Laspeyres market-value index with a divisor; liquidity rule; number of constituents by the AIC, smoothed over six months
- Nine types of crypto indices share the same backbone and differ in universe, weights, data and rules
- Same ten coins, three weightings: Bitcoin 80.8%, 50.7% or 10% of the index; a protection floor cut the drawdown from -76.6% to -13.9%
- Our ten-coin replica tracks CRIX with correlation 0.990 and tracking error 9.3% a year; Bitcoin alone: 0.971 and 15.6%
- CRIX 2018–2026: CAGR 24.0%, volatility 65.0%, maximum drawdown -77.4%; record 21 523.34 on 6 October 2025
- VCRIX follows the 30-day realised volatility (correlation 0.94); GARCH persistence 0.998
- A spot Bitcoin fund: tracking difference -0.28% a year, tracking error 3.5% when prices are synchronous

Check Yourself: Quiz (1)

1. On a rebalancing day the market value of the new basket is 2 045.41 billion USD and the index stands at 9 521.15. The new divisor is
   (a) 9 521.15 (b) 214.83 million (c) 2 045.41 billion (d) 0.0047
2. $\mathrm{AIC}(k)$: -946.2, -953.3, -953.6, -952.3 for $k = 2, 3, 4, 5$. The first-rise rule chooses
   (a) $k = 2$ (b) $k = 3$ (c) $k = 4$ (d) $k = 5$
3. An index weights each coin by the square root of its market value. Compared with market-value weights it
   (a) changes the ranking of coins (b) gives more weight to smaller coins (c) removes Bitcoin (d) fixes the number of coins at ten

⊡ Answers: 1 (b), 2 (c), 3 (b); more questions in the practice quiz of Lecture 3 on the course website

Check Yourself: Quiz (2)

4. A fund has a tracking difference of -0.28% a year and a tracking error of 3.5%. Which statement is right?
   (a) it lags its index by about 3.5% every year (b) it lags by about 0.28% a year on average and deviates by about 3.5% a year around that
5. VCRIX jumps on 13 March 2020 and drops sharply 30 days later, on 12 April 2020. The drop is explained by
   (a) a calm day on 12 April (b) the crash day leaving the 30-day window
6. The ex-post top 5 of 2026, bought in 2018, earned 17.6% a year; the ex-ante top 5 earned 7.1%. The difference measures
   (a) the skill of the index provider (b) look-ahead and survivorship bias

⊡ Answers: 4 (b), 5 (b), 6 (b)

Interpretation Exercises

1. Bitcoin holds 82.2% of the value of ten coins; $1/\mathrm{HHI} = 1.44$. A colleague proposes an equally weighted index "to represent the market better". Comment on representativeness and investibility.
2. The AIC curve of the last quarter falls by 284 points from $k = 1$ to $k = 2$ and by 0.3 from $k = 3$ to $k = 4$. What does this say about the number of constituents and about the stability of the choice? Why smooth the curve over six months?
3. Our replica: correlation 0.990 and tracking error 9.3% against CRIX; capped below 50%: 0.979 and 14.5%. What do these numbers say about the published index? What do they not prove?
4. CRIX: CAGR 24.0%, volatility 65.0%; Bitcoin: 29.4% and 61.7% (2018–2026). Would a broad crypto index have helped a Bitcoin investor? In which year was the answer different?
5. A Bitcoin-and-cash index with a protection floor: CAGR 9.9%, maximum drawdown −13.9%; Bitcoin: 32.3% and −76.6%. For which investor is the floor index the better benchmark? What does the investor give up?
6. IBIT: tracking difference −0.28% a year; tracking error 3.5% at 16:00 New York and 20.6% at midnight UTC. Which number would you report to an investor, and why?

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- Index providers and data
  - Royalton CRIX and VCRIX: thecrix.de (methodology and daily values, accessed 7 November 2026); Royalton Partners: CRIX
  - Bloomberg (2020): Bloomberg Galaxy Crypto Index methodology, December 2020
  - CoinDesk Indices (2025): CoinDesk 20 Index methodology
  - Nasdaq: NCI; S&P Dow Jones Indices: Cryptocurrency Indices
  - iShares: IBIT (sponsor fee); SEC (2024) (approval of spot Bitcoin ETFs)
  - Bitwise (2025); CoinDesk (2024); MarketVector (2025a); CoinDesk (2025); Volmex; LIXX (2023); MarketVector (2025b); MarketVector (2024); CoinMarketCap (2025)
  - FRED: DFF: effective federal funds rate, daily
  - Coin Metrics Community API (coin supply); EODHD (daily and hourly prices)

Appendix: Data Notes

- CRIX and VCRIX were read from the chart data of thecrix.de on 7 November 2026
  - 3 321 daily CRIX values, 6 September 2017 – 6 November 2026; 28 missing days in 2026 (22 March, 14 May – 7 June, 31 August, 5 November)
  - returns are computed only between consecutive days; a return over a gap is not used
- Before 16 March 2018 CRIX moves one day ahead of the EODHD Bitcoin close (correlation of daily returns 0.90 with the next day's Bitcoin return in 2017, 0.06 with the same day); thecrix.de notes a different data source for 2016–2018
- Market values of the replica: EODHD close × Coin Metrics current supply (`SplyCur`); volumes: EODHD, USD
- The replica is a teaching replication of the published rules on a small universe, not the official index

Appendix: Kernel Density and Leave-One-Out

- Kernel density estimate of a sample $x_1, \dots, x_n$ with kernel K and bandwidth h:

$$\hat{f}(x) = \frac{1}{nh} \sum_{s=1}^{n} K\Big(\frac{x - x_s}{h}\Big), \qquad K(u) = \varphi(u) = \frac{e^{-u^2/2}}{\sqrt{2\pi}}$$

- Evaluating $\hat{f}$ at its own data overstates the likelihood: each point sits on its own bump
- **Leave-one-out**: drop the point itself, $\hat{f}_{-t}(x_t) = \frac{1}{(n-1)h} \sum_{s \neq t} K\big((x_t - x_s)/h\big)$
- Bandwidth (Silverman's rule): $h = 0.9 \, \min(\hat{\sigma}, \widehat{\mathrm{IQR}}/1.349) \, n^{-1/5}$
  - last quarter of the replica: $n = 90$ residuals, $h = 0.00069$ in log-return units (0.069 percentage points)